

Piyush Hinduja

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EDUCATION

Master of Science in Computer Science

University of Utah, USA

GPA: **3.91**/4.0

2023 – 2025

Bachelor of Engineering in Computer Engineering

University of Mumbai, India

GPA: **3.85**/4.0

2019 – 2023

SKILLS

Languages: *Expert (5+ yrs):* Python, Java, JavaScript *Intermediate (3+ yrs):* C, C++ *Beginner (1+ yrs):* PHP, Kotlin, SQL, Shell

AI/ML: LLMs, PyTorch, HuggingFace, Langchain, TensorFlow, Scikit-Learn, EasyOCR, spaCy

Data Processing: RESTful APIs, Pandas, PySpark, Databricks, OCRs, Matplotlib, Seaborn, DuckDB, PowerBI, Tableau

Applications and Databases: FastAPI, Streamlit, Bash, Preswald, MongoDB, MySQL, Amazon RDS, Firebase, ChromaDB

Cloud & Miscellaneous: AWS (S3, SageMaker, Textract, EC2), Azure, Docker, Kubernetes, Git, CI/CD, Agile/SCRUM, JIRA

WORK EXPERIENCE

AI Engineer (formerly Data Analyst)

University of Utah

March 2024 – Present

Salt Lake City, Utah

- Constructed staged **LLM-ETL pipelines** for targeted text extraction and insight generation across a large corpus of financial documents (from SEC.gov and EDGAR), enabling **90% faster**, context-aware information retrieval
- Performed deep NLP analysis using **encoder-only models like DeBERTa & FinBERT (PyTorch, Transformers)** for financial document processing, **improving the prediction accuracy from 66% to 94.13%**
- Developed a **Regex + Fuzzy Match system achieving 99.6% extraction accuracy**, ensuring precise entity identification and section-level text matching across heterogeneous filings
- Built robust **prompt engineering** solutions to integrate LLMs & GenAIs (OpenAI, Anthropic, and Gemini) for fast and automated text analysis

Software Developer

PONO.AI

September 2025 – Present

Salt Lake City, Utah

- Implemented backend services for **voice AI-based customer onboarding**, enabling users to complete in-depth onboarding via a 20-minute **AI-driven Twilio call** instead of manual form entry, **increasing customer engagement and onboarding completion rates by 40%**
- Developed **MCP agents** integrated with web and company databases to fetch contextual information dynamically, **improving response relevance and overall user experience**

AI Engineer

Giant Analytics

July 2025 – January 2026

Remote

- Pre-trained **decoder-only models, like LLaMA 3.1**, on Indian Law data extracted via Indian Kanoon and Bharat Law APIs for domain-specific legal reasoning using **AWS SageMaker and S3**
- Integrated the model into a **RAG-based SaaS platform** that retrieves and cites relevant cases during chat; projected to boost answer accuracy from **56% to 93%**, significantly **outperforming generic OpenAI & Gemini models**

Software Developer

Squeeze Media LLC

May 2025 – June 2025

Orem, Utah

- Built an AI-powered warm transfer system for inbound/outbound **Twilio calls using FastAPI, Deepgram SST, and WebSocket** streaming
- Boosted company sales by 28%** by ensuring 60% more complete calls without drop-off or repeated queries

PROJECTS

SEC Comment Letter Analysis | LLMs, OCR, Claude API

- Engineered a rolling-window algorithm to process and bundle SEC comment letter conversations, analyzing data from **50,000+ companies** via the EDGAR database
- Designed an **OCR pipeline using the EasyOCR library** to extract text from PDFs with **95% accuracy**, processing 2-page documents in under 2 seconds
- Constructed prompt-based labeling system using **Claude API** to classify letter content with **98.25% accuracy**, integrating metadata for contextual inference

Road Damage Detection | Object-Detection, YOLOv5, FasterRCNN

[GitHub Link](#)

- Collaborated with a team of four to develop a road damage detection system using **YOLOv5 and FasterRCNN**, integrating a user-friendly web interface (deployed via **Streamlit**), leveraging the RDD-2020 dataset
- Achieved model accuracies of **52.67% with YOLOv5 and 44.45% with FasterRCNN** by training and fine-tuning on different hyper-parameters
- The system plays a vital role, delivering a **70% improvement in road damage detection accuracy** over manual methods, helping lower the frequency of accidents and fatalities

Wireless Spectrum Analysis Research | Python, Linux Bash, NVIDIA/CUDA

- Engineered automated deployment system using Python and Linux Bash to streamline **NVIDIA/CUDA toolkit** integration on high-performance computing clusters, **reducing overhead by 90%**
- Analyzed wireless spectrum data to uncover scientific insights and **optimize power distribution across 20+ campus routing devices**

Dependency Parser | PyTorch, GloVe Embeddings

[GitHub Link](#)

- Architected an end-to-end **Transition-Based Dependency Parser in PyTorch**, implementing a multi-class classifier for predicting parsing actions based on stack and buffer states
- Designed modular pipeline incorporating data preprocessing, model architecture, training workflow, and evaluation metrics
- Integrated **GloVe word embeddings** for enhanced semantic representation, contributing to **UAS of 0.76 and LAS of 0.705**